

popquiz cosmology

 This is a preview of the published version of the quiz

Started: Mar 22 at 7:21pm

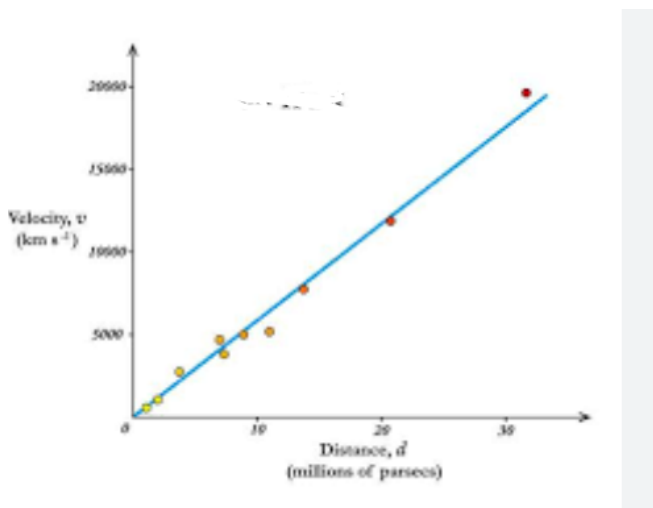
Quiz Instructions

Based on previous lecture



Question 1 10 pts

In class we discussed this graph:



On the x-axis you the distance of some galaxies. On the y-axis the relative speed of the galaxies.

More distant galaxies (based on 1920s understanding)

- ☐ moves faster relative to us then closer galaxies
- ☐ moves slower relative to us than closer galaxies
- ☐ all the galaxies move at the same speed relative to us
- ☐ we live in a static universe.



Question 2 10 pts

Building on the previous question. The graph shows a linear relationship between speed and distance. this relationship is called



Hubble's law



Einstein's law



Zwicky's law



Jansky's law



Question 3 10 pts

Who introduced an "anti gravity" force called the lambda factor. To support the idea of a static Universe. (at rest).



Albert Einstein



Edwin Hubble



Georges Lemaitre



George Gamow



Question 4 10 pts

now we know that the Universe



is accelerating



is static



is slowing down and will experience a big crunch



is oscillating



Question 5 10 pts

The spectra of distant galaxies are



redshifted



blueshifted



same as in lab



I don't pay attention and I don't go to class



Question 6 10 pts

What is a gravitational wave, in the sense used in general relativity?



A traveling ripple in the curvature of spacetime produced by accelerating masses



A type of electromagnetic radiation similar to visible light



C A static distortion of space that never changes with time



A stream of particles that carry gravity through space



Question 7 5 pts

Who predicted the cosmic microwave background but didn't get a shared nobel prize when observed?



George Gamow and colleagues



Arno Penzias and colleagues



Albert Einstein and Colleagues



Edwin Hubble and Colleagues



Question 8 5 pts

In a famous experiment the CMB was discovered with a huge antenna. But first they had to get rid of pigeons. The pigeon droppings were a issue. how did they get rid of them?



shot gun



they sent them away as far as they could



they ate them using a French recipe. Pigeons au Vinaigre d'Échalote.



No sense. birds do not nest in antenna.



Question 9 10 pts

During the cold war. The US first detected ----- in the late 1960s with Vela satellites that were built to monitor nuclear tests after the Nuclear Test Ban Treaty

- ☐ gamma ray bursts
- ☐ radio waves bursts
- ☐ UV flashes
- ☐ gravitational waves



Question 10 10 pts

Pictures of BH

- ☐ are taken using radiowaves
- ☐ are taken using visible light
- ☐ were never taken
- ☐ using IR



Question 11 10 pts

Why is the cosmic microwave background (CMB) often described as a “snapshot” of the early universe?

- ☐ It is radiation released when the universe became transparent to light
- ☐ It records light from the very first stars that formed
- ☐ It shows galaxies exactly as they look today
- ☐ It is light produced by current star formation in our galaxy

Not saved

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